

High Blood Pressure Risk Factors in Bangladesh

Statistical Analysis of BDHS 2022

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Tools: IBM SPSS Statistics

Dataset: Bangladesh Demographic and Health Survey 2022 (N = 10,086)

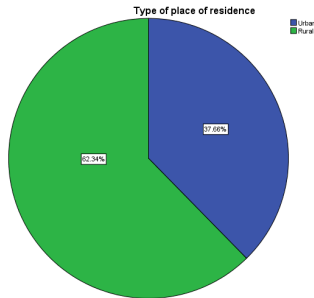
Project Overview

This Project analyses the prevalence and causes of hypertension in Bangladesh, using data from the Bangladesh Demographic and Health Survey (BDHS) 2022. Both inferential and descriptive statistical techniques were used using SPSS. It mainly investigates how demographic, socioeconomic, and lifestyle factors affect high blood pressure.

Detail	Value
Dataset	BDHS 2022
Sample size	10,086 participants
Divisions covered	8
Variables analyzed	11
Statistical tests used	Chi-square, Pearson r, Linear Regression
HBP prevalence found	17.5%
Software	IBM SPSS Statistics

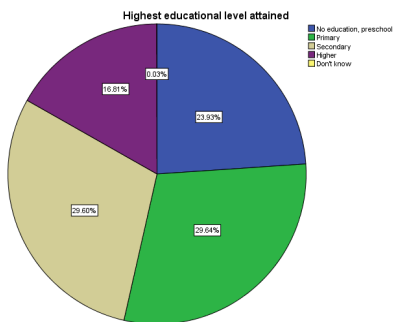
Who Was Studied?

1. Residence type (Urban/Rural)



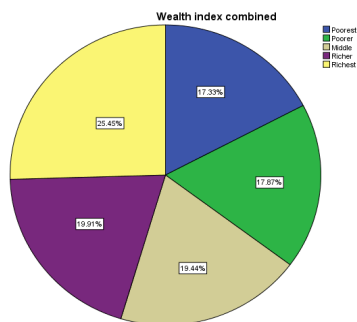
62.3% of participants were from rural areas, reflecting Bangladesh's predominantly rural population.

2. Education Level



Primary and secondary education were equally represented at 29.6% each

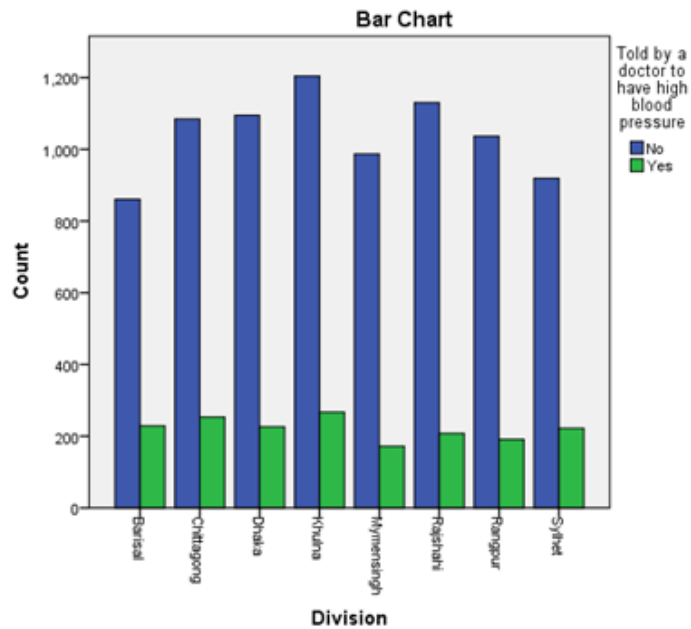
3. Wealth Index



The wealthiest quintile was the largest group at 25.5%.

Chi-Square Findings

1. Division and High Blood Pressure



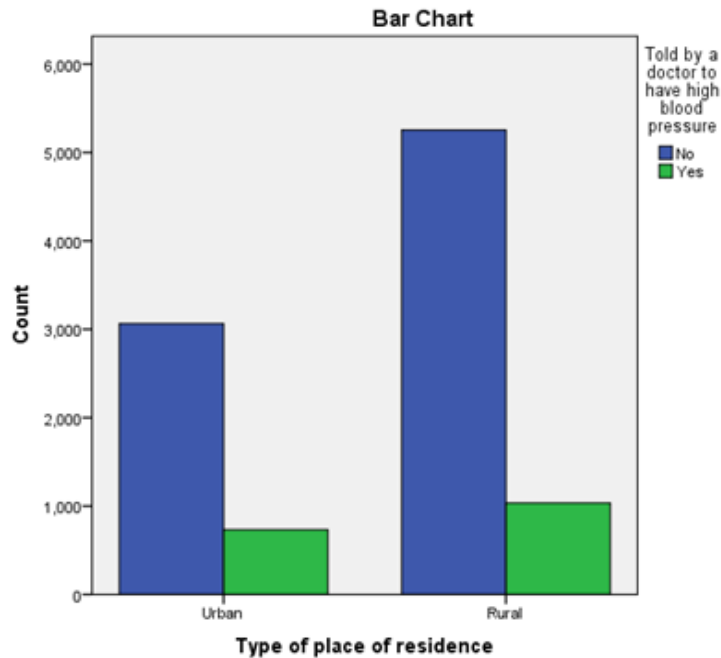
The bar chart shows that in all divisions, the number of respondents without high blood pressure is much higher than those diagnosed with it. Overall, 82.5% of the 10,086 respondents reported not having high blood pressure, while 17.5% reported having it. Barisal and Sylhet had the highest proportions of high blood pressure, whereas Mymensingh had the lowest. Although Khulna reported the highest number of cases, this may be due to its larger sample size.

Statistical Test Result:

Test	Value	p-value	Conclusion
Pearson Chi-Square	27.133	< 0.000	Significant association

Key Insight: Barisal (21%) and Sylhet (19.5%) have the highest HBP rates.

2. Residence and High Blood Pressure



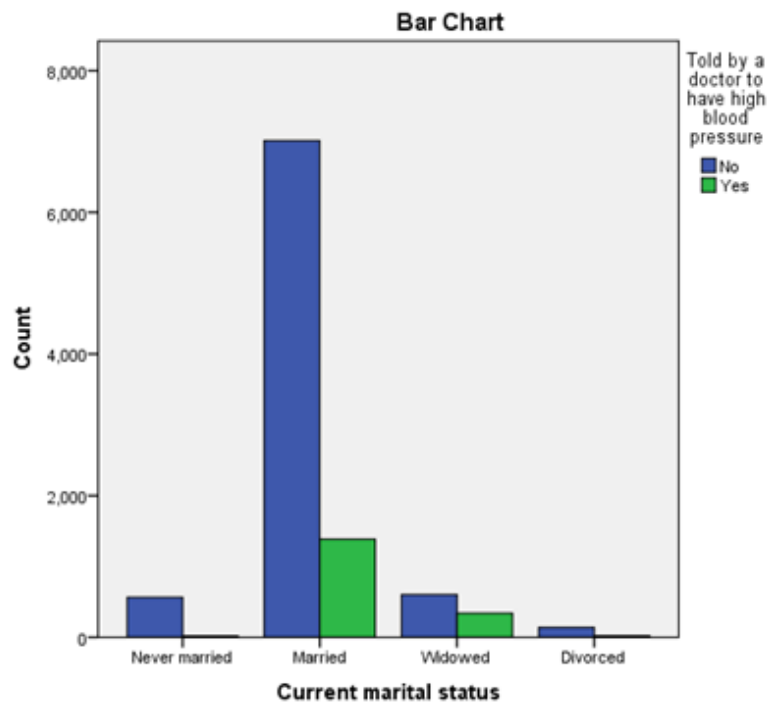
In the study, the bar chart shows differences in high blood pressure between urban and rural residents. Although the total number of cases is higher in rural areas due to a larger population size, the prevalence is higher in urban residents (19.3%) compared to rural residents (16.4%). Among the 1,768 individuals with high blood pressure, 58.5% are from rural areas and 41.5% are from urban areas, which reflects the larger rural sample size rather than a higher risk.

Statistical Test Result:

Test	Value	p-value	Conclusion
Pearson Chi-Square	13.604	< 0.000	Significant association

Key Insights: Urban residents have higher HBP prevalence (19.3%) than rural residents (16.4%).

3. Current Marital Status and High Blood Pressure



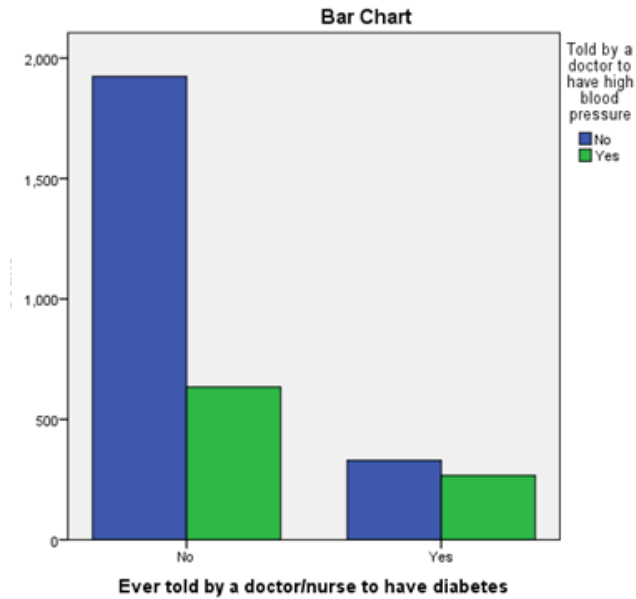
In the study, high blood pressure prevalence differs by marital status. Widowed individuals show the highest prevalence (36.1%), followed by married (16.5%) and divorced (13.8%), while never-married individuals have the lowest (3.3%). Although married individuals make up the largest proportion of the sample (8,400 out of 10,086), resulting in the highest number of cases (78.5%), this reflects the sample size rather than risk level. Widowed individuals account for 19.2% of cases despite their smaller group size.

Statistical Test Result:

Test	Value	p-value	Conclusion
Pearson Chi-Square	313.594	< 0.000	Significant association

Key Insight: Widowed individuals face the highest risk at 36.1% — likely age-related

4. Diabetes and High Blood Pressure



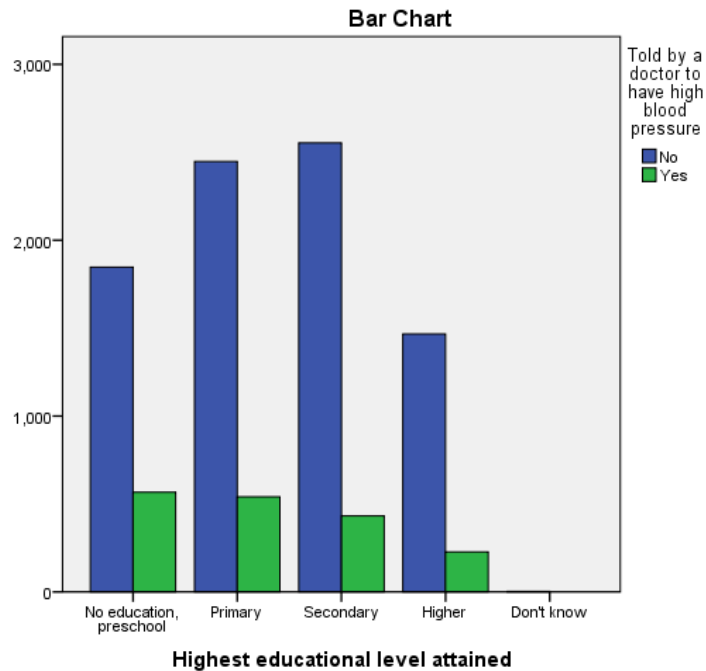
In the study, the results show a significant association between diabetes and high blood pressure. Among individuals with diabetes, 44.7% have high blood pressure, compared to 24.8% among those without diabetes, indicating a higher prevalence in diabetic individuals. Out of the 901 individuals with high blood pressure, 70.4% are non-diabetic, and 29.6% are diabetic. However, this distribution is influenced by the larger proportion of non-diabetic participants in the sample, while diabetics represent 18.9% of the total population. The Chi-square test confirms a statistically significant association between diabetes status and high blood pressure.

Statistical Test Result:

Test	Value	p-value	Conclusion
Pearson Chi-Square	94.316	< 0.000	Significant association

Key Insight: Diabetics are nearly twice as likely to have HBP (44.7% vs 24.8%)

5. Education and High Blood Pressure



The study examined the relationship between high blood pressure and education level. The prevalence of high blood pressure was highest among individuals with no education or only preschool education (23.5%), followed by primary education (18.1%), secondary education (14.5%), and lowest among higher educated participants (13.5%). Overall, the results show a decreasing trend in high blood pressure prevalence with increasing education level. This suggests that lower educational attainment is associated with a higher likelihood of having high blood pressure.

Statistical Test Result:

Test	Value	p-value	Conclusion
Pearson Chi-Square	99.393	< 0.000	Significant association

Key Insight: Lower education = higher HBP risk (23.5% → 13.5% gradient)

6. Wealth Index and High Blood Pressure



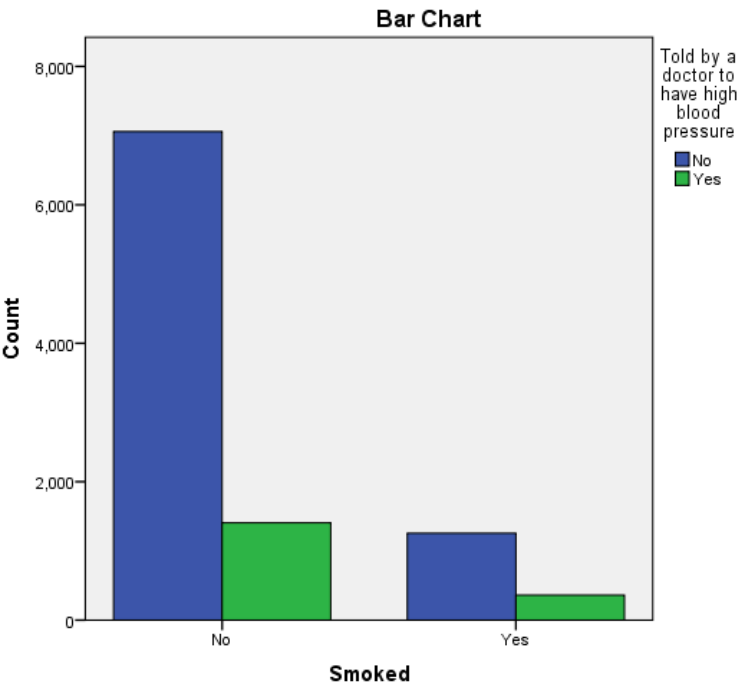
The bar graph shows differences in high blood pressure across wealth categories. In the poorest group, 12.5% reported high blood pressure while 87.5% did not. This increases slightly in the poorer group, where 14.4% had high blood pressure and 85.6% did not. In the middle group, 17.0% reported high blood pressure compared to 83.0% without it. The wealthier group shows 18.9% with high blood pressure, and the highest prevalence is observed in the wealthiest group at 22.5%. Overall, the results indicate an increasing trend in high blood pressure with higher wealth status, suggesting higher prevalence among wealthier groups.

Statistical Test Result:

Test	Value	p-value	Conclusion
Pearson Chi-Square	88.983	< 0.000	Significant association

Key Insight: The wealthiest group has the highest HBP (22.5%) — better diagnosis access.

7. Smoking and High Blood Pressure



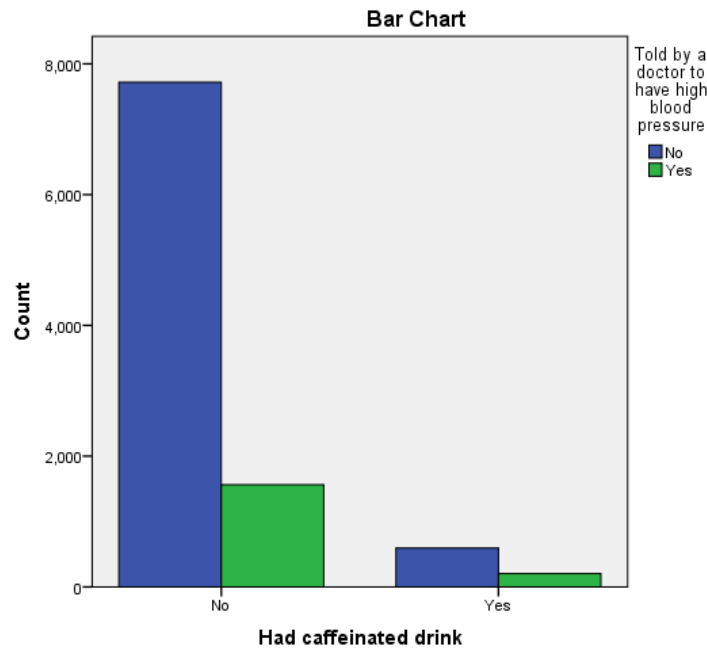
The bar chart shows the relationship between smoking status and high blood pressure. Among non-smokers, 16.6% reported having high blood pressure, while 83.4% did not. Among smokers, 22.4% had high blood pressure and 77.6% did not. Overall, the prevalence of high blood pressure is higher among smokers compared to non-smokers, suggesting an association between smoking and increased risk of high blood pressure.

Statistical Test Result:

Test	Value	p-value	Conclusion
Pearson Chi-Square	31.446	< 0.000	Significant association

Key Insight: Smokers show 35% higher HBP rate than non-smokers

8. Caffeinated Drinks and High Blood Pressure



The bar chart shows differences in high blood pressure based on caffeinated beverage consumption. Among participants who did not consume caffeinated drinks, 16.8% had high blood pressure, while 83.2% did not. In comparison, 25.5% of those who consumed caffeinated beverages had high blood pressure, while 74.5% did not. Overall, the results indicate a higher prevalence of high blood pressure among caffeinated drink consumers, suggesting a possible association between caffeine intake and elevated blood pressure, although the difference between groups is not large.

Statistical Test Result:

Test	Value	p-value	Conclusion
Pearson Chi-Square	31.446	< 0.000	Significant association

Key Insight: Caffeine consumers show notably higher HBP prevalence (25.5%)

Does Age Predict Blood Pressure?

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	906220.051	1	906220.051	1053.126	.000 ^b
	Residual	8677330.958	10084	860.505		
	Total	9583551.009	10085			

a. Dependent Variable: Systolic blood pressure measurement

b. Predictors: (Constant), Age of household members

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	103.059	.793		129.935	.000	101.505	104.614
	Age of household members	.582	.018	.308	32.452	.000	.546	.617

a. Dependent Variable: Systolic blood pressure measurement

Correlation Finding:

A Pearson correlation analysis revealed a weak-to-moderate positive relationship between age and systolic blood pressure ($r = 0.308$, $p < 0.001$). This means that as age increases, blood pressure tends to rise — but the relationship is modest.

Regression Finding:

Simple linear regression confirmed age as a statistically significant predictor of systolic BP ($F = 1053.1$, $p < 0.001$). For every additional year of age, systolic BP increases by approximately 0.582 mmHg. However, age alone explains only 9.5% of BP variation ($R^2 = 0.095$), meaning lifestyle factors, health conditions, and socioeconomic status account for the remaining 90.5%.

Data Quality Note:

One extreme outlier (996 mmHg) was identified in the dataset — a clear data entry error. This inflates the mean and skewness statistics and should be removed before any clinical modelling.

What Do These Findings Mean?

These findings show how high blood pressure varies across different population groups and lifestyle factors such as division, residence, marital status, education, wealth, diabetes, smoking, and caffeine consumption.

Overall, the results consistently suggest that high blood pressure is not randomly distributed in the population. Instead, it is associated with several demographic and behavioral factors. In many cases, certain groups show higher prevalence compared to others, indicating potential risk patterns.

The Chi-square tests in the analysis confirm that these associations are statistically significant, meaning the relationships observed are unlikely to be due to chance. However, in some variables, differences in case numbers may also reflect sample size distribution rather than true risk differences.

Summary Table:

Risk Factor	HBP Prevalence	Public Health Priority
Diabetes comorbidity	44.7%	Integrated diabetes–HBP care
Widowed individuals	36.1%	Elderly screening programs
Caffeine consumption	25.5%	Lifestyle counselling
No formal education	23.5%	Health literacy campaigns
Smokers	22.4%	Smoking cessation programs
Wealthiest quintile	22.5%	Urban NCD prevention
National average	17.5%	Baseline for comparison

In summary, the findings indicate that high blood pressure is influenced by multiple social and health-related factors, highlighting the importance of targeted public health awareness and prevention strategies.

